



OPEN Frequency ratio model for flood susceptibility mapping in Nigeria

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Annually, the low-lying Niger Delta region of Nigeria, one of the country's most flood-prone areas, experiences severe flooding that causes extensive damage to infrastructure, ecosystems, and socio-economic stability. Hence, accurate flood susceptibility assessment and risk mapping are crucial for supporting effective mitigation and planning, which is the focus of this study. To address this, an integrated Geographic Information Systems (GIS) and Frequency Ratio (FR) approach is used to delineate flood hazard zones and identify high-risk areas across the region. Notably, this represents the first application of the FR model across the unified Niger Delta, encompassing its six core states. The lack of documented flood-susceptibility assessments in this economically vital region underscores a significant research gap. Ten multicollinearity-free flood-conditioning factors, including rainfall, distance to river, drainage density, land use/land cover, elevation, slope, NDVI, soil type, curvature, and topographic wetness index, were analysed to generate a flood susceptibility map classified into five levels. The results indicate that 71.79% (26,297.21 Km²) of the region falls within the moderate to very high susceptibility category. In comparison, 28.21% (10,335.46 Km²) exhibits low to very low susceptibility, with Bayelsa, Rivers, and Delta States identified as the most at risk. The model achieved a reliability rate of 83.16% based on the ROC–AUC analysis, confirming its predictive accuracy. Overall, the findings provide critical, data-driven insights for policymakers and urban planners, supporting a shift from reactive disaster response to proactive flood risk management, and offering a transferable, robust framework for flood mitigation and sustainable development in similar flood-prone regions.

Keywords Flood susceptibility, Disaster, Geographic information system, Frequency ratio model, Risk management

The devastating impact of floods, one of the most critical natural threats worldwide, creates immense challenges for human populations^{1,2}. These events lead to a range of serious outcomes, including loss of life and injury, alongside extensive damage to infrastructure, economic systems, and social structures³. Flood events have extensive impacts on both physical and mental health. Following a flood, individuals often suffer from psychological distress, including anxiety, depression, Post-Traumatic Stress Disorder (PTSD), psychosis, and insomnia⁴. Globally, 89% of people exposed to flood risk live in low- and middle-income countries, highlighting the heightened vulnerability of the developing world^{5,6}. The severity of flooding is increasingly amplified by the effects of climate change, haphazard urbanisation, and ineffective land management practices^{7,8}. Due to its significant socio-economic repercussions, the hazard has become a major focus of study worldwide.

Nigeria is divided into six geopolitical zones, and the South-South zone explicitly includes all six states that constitute the core of the Niger Delta. Therefore, the term “Niger Delta” is used throughout this study. This region experiences perennial flooding that poses significant challenges to its communities and development^{9,10} and it is particularly prone to flooding due to heavy rainfall¹¹. In 2022, one of the most devastating floods occurred in Nigeria, and 600 people were reported dead from this event^{12,13}. This same year, Bayelsa State, within the Niger Delta region, saw 300 communities affected by floods, impacting over 1.3 million people¹⁴. The recurrent

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nature of floods and their devastating effects highlight the urgent need for effective mitigation and management strategies in the Niger Delta region¹⁵. Hence, there is a need for flood modelling to predict susceptible areas and guide effective adaptation and mitigation measures.

Literature review

Flood conditioning factors

Flood susceptibility mapping is crucial for effective disaster management, vulnerability assessment, and future preparedness^{16,17}. The selection of appropriate flood conditioning factors is therefore paramount to the accuracy and reliability of flood susceptibility models, as these factors reflect the environmental, topographical, and anthropogenic influences on flood generation and propagation¹⁸. The chosen factors for this study which include: rainfall, distance to river, drainage density, land use/land cover (LULC), elevation, slope, Normalized Difference Vegetation Index (NDVI), soil type, curvature, and Topographic Wetness Index (TWI), are widely recognized and empirically supported in the literature for flood susceptibility mapping using the Frequency Ratio (FR) model and other geospatial techniques^{18,19}. As a direct hydrological input, rainfall is a primary trigger for floods. High rainfall intensity and duration directly correlate with increased runoff and inundation, making it a critical factor in flood susceptibility assessments. Studies have consistently shown that rainfall is a significant contributor to flooding in Nigeria^{20,21}. Proximity to river networks significantly influences an area's susceptibility to flooding. Areas closer to rivers are inherently more prone to inundation due to overbank flow, especially in regions with complex fluvial environments like the Niger Delta²². Drainage density is a morphometric parameter that reflects the efficiency of surface runoff. Higher drainage density typically indicates a quicker response to rainfall events and an increased likelihood of flooding, as water is transported rapidly through the landscape²¹. LULC patterns critically affect infiltration, runoff, and storage capacity. Urbanisation, characterised by impervious surfaces, reduces infiltration and accelerates runoff, while deforestation can exacerbate erosion and increase flood volumes. Land use has also been identified as a major contributing factor to flooding²¹. Elevation also plays a fundamental role, as lower elevation areas are naturally more susceptible to water accumulation. The relative flatness of areas like the Niger Delta contributes to its vulnerability²³. Gentle slopes generally lead to slower runoff and increased water accumulation, making them more susceptible to flooding than steep slopes, where water drains more quickly¹⁹. NDVI serves as an indicator of vegetation cover density, which influences rainfall interception, evapotranspiration, and soil moisture. Denser vegetation can reduce runoff and erosion, thus mitigating flood impacts. Conversely, decreasing vegetation cover is associated with higher flood sensitivity²⁴. Soil characteristics, such as permeability and infiltration rates, determine how much water the ground can absorb. Soils with low permeability contribute to higher surface runoff and increased flood risk²⁵. Curvature influences surface flow paths and accumulation patterns. Concave areas tend to collect water, increasing flood susceptibility²⁶. TWI quantifies the tendency of an area to accumulate water based on local slope and upslope contributing area. High TWI values indicate areas prone to saturation and increased flood risk¹⁹. Each of these selected factors, when integrated, provides a comprehensive and spatially explicit understanding of the complex interplay of environmental conditions that drive flood susceptibility within the study area, making them indispensable for accurate flood risk assessment and mitigation strategies.

Geographical context: The Niger Delta region, Nigeria

The Niger Delta, like other coastal regions, is highly susceptible to the impacts of climate change, which manifests through increased rainfall intensity and rising sea levels²⁷. Flood events in Nigeria have recently become more challenging, and coastal regions, including the Niger Delta, are predicted to become increasingly susceptible to flooding due to climate change²⁸. The geomorphology of the Niger Delta, characterised by its low-lying coastal plains and intricate river networks, naturally predisposes it to flooding²². Hydrological processes in the Niger Delta are complex and influenced by factors such as the evolution of flood flows, human activities, and the presence of numerous river systems. The region experiences perennial flooding, prompting evaluations of flood risk mitigation strategies²⁷.

Methodological justification: the frequency ratio (FR) model

Traditionally, physics-based flood models have been used for flood mapping, but these are costly to calibrate and often lack generalizability across different river basins due to their sensitivity to site-specific parameters and human-regulated infrastructure²⁹. This limitation has spurred the development of Geographic Information System (GIS)-based techniques, statistical, machine learning (ML) and other related approaches to improve flood prediction accuracy and applicability³⁰. While the FR model is utilised in this study because of its proven effectiveness as a robust bivariate statistical method, it is crucial to position this research within the broader advancements in geospatial analysis and flood modelling.

Principles and effectiveness of the FR model

The underlying principles of FR, which involve quantifying the relationship between flood occurrences and conditioning factors, are fundamental to even the most sophisticated ML frameworks, thereby establishing a significant scientific continuity. The FR model statistically relates flood inventory data to various thematic layers, effectively identifying patterns from observed flood events. This methodology is conceptually analogous to how many ML algorithms learn relationships and generate predictions³¹. Consequently, while FR is not an ML algorithm, it provides a transparent and interpretable statistical basis that informs more complex ML approaches³². Several studies, in Malaysia, Korea, China, India, Pakistan, Thailand and Nigeria, have utilised the bivariate statistical FR model^{18,33–38} to map flood vulnerability. To assess the contribution of different factor classes to flood events, the FR method calculates the ratio of flooding probability to non-flooding probability for a given attribute³⁹. The efficacy of the FR model is well-established, with prior studies reporting success

rates of 64–96%³⁴. For instance, a study in Kota Belud, Sabah, Malaysia, used the FR model to generate a flood susceptibility map with five classes, and the model's accuracy was validated with an Area Under Curve (AUC) of 0.82⁴⁰. Similarly, a study in the Swat River Basin, Pakistan, using the FR model, showed an AUC of 90.3%³⁸. Another study in the Upper Krishna Basin, India, reported a 68% AUC rate for the FR model⁴¹. Collectively, these studies demonstrate the robustness and predictive capability of the FR model in flood susceptibility mapping, thereby justifying its application in the present study.

Application of the FR model in Nigeria

While the FR model is a powerful tool, its application for comprehensive flood risk assessment in Nigeria, particularly in the Niger Delta region, remains limited. A notable study by Efiogbe, et al.¹⁹ applied this methodology to the Calabar Metropolis in Cross River State, integrating FR with GIS to create a flood susceptibility map. Their findings classified the area's risk levels, showing that 9.37% was very highly susceptible, 19.97% was high, and 31.79% was at moderate risk. The remaining area was predominantly low-risk (39%), with only 10.59% being very low risk. Model validation using the Receiver Operating Characteristic (ROC) curve yielded an AUC value of 74.81%, indicating an acceptable level of predictive accuracy. A further application of the FR model is illustrated in the work of Isiaka, et al.¹⁸. While their study was conducted in Nigeria, its scope was outside the Niger Delta region. In a focused application on Lagos State, the nation's economic capital, the GIS-based FR model identified 12.54% of the area as being at very high flood risk, while only 11.62% was deemed very low risk. Validation results confirmed satisfactory model performance, with success and Prediction Rates (PR) of 64% and 61%, respectively, based on AUC analysis.

FR model in data-scarce regions and future integration

In regions where comprehensive, high-resolution, and temporally consistent data essential for advanced ML training are scarce, the FR model offers a practical and reliable solution³¹. It effectively leverages readily available existing datasets, such as historical flood records and environmental data, to produce meaningful flood susceptibility maps^{42,43}. This approach demonstrates that robust scientific inquiry and valuable insights can be achieved despite data limitations, making the FR model a highly relevant tool in such contexts. By quantifying how different factors contribute to flooding, the FR model offers crucial information on flood vulnerability, thus allowing for well-targeted strategies to mitigate disaster risk.

Even with a focus on the FR model, this study aligns with contemporary data acquisition trends by utilising readily available and widely recognised geospatial data sources. For example, the Digital Elevation Model (DEM) data was obtained from the USGS Earth Explorer, a platform that provides access to a wealth of publicly available satellite imagery and topographic data. Similarly, the LULC and NDVI data were derived from Sentinel-2A imagery, showcasing the integration of current satellite data for crucial environmental parameters. This indicates that the foundational data sources and processing techniques share commonalities with modern geospatial workflows, ensuring the scientific relevance and comparability of the input data³¹.

As data availability improves, the flood susceptibility map generated using an FR model can serve as a baseline for future research, as the frequency ratios themselves could be used as input features in more complex ML models, effectively integrating statistical insights from FR into a predictive ML framework^{31,44}. Furthermore, the insights gained from applying FR in data-scarce regions like Nigeria can inform and guide the application of more advanced techniques in similar global contexts, fostering broader scientific continuity in flood modelling research.

Addressing methodological gaps and research significance

A significant issue in existing flood assessment studies in Nigeria is the infrequent conduction of multicollinearity assessments when selecting influencing factors, and this can be problematic because it can produce misleading model results by inflating the variance of coefficient estimates and undermining the reliability of the conclusion⁴⁵. Therefore, evaluating multicollinearity is an important step in ensuring model robustness. Also, flood risk analysis is rarely done at the regional scale, often due to data limitations and the complexity involved in integrating diverse factors across larger areas. While computational capacities are improving, upscaling sophisticated methodologies to broader geographical scales, especially for data-sparse environments, remains a challenge⁴⁶. Furthermore, the Niger Delta region of Nigeria, though not the most severely affected by floods compared to other regions, is also significantly impacted by flooding due to its low-lying coastal areas, extensive river networks, and heavy seasonal rainfall¹³. In-depth flood research in this region is scarce, limiting data and understanding of its flood patterns and socio-economic effects. This research gap is particularly concerning given the area's contribution of a large share of Nigeria's oil and gas resources, establishing its national economic value⁴⁷. Floods in this economically vital zone risk disrupting production, damaging infrastructure, and harming local communities.

This study aims to address these identified gaps by leveraging the robust capabilities of the FR model in the Niger Delta region of Nigeria. A key methodological step in this study was the systematic evaluation of multicollinearity among the flood conditioning factors. This was done to ensure that the final model retained only independent and significant variables^{48,49}. This enhances the model's robustness and the credibility of its findings, which in turn supports more effective flood risk management decisions. Also, by quantifying how different factors contribute to flooding, the FR model offers crucial information on flood susceptibility across the region, thus allowing for well-targeted strategies to mitigate disaster risk. Furthermore, as in-depth flood research in the Niger Delta region remains limited, this study helps bridge this knowledge gap by providing essential data to protect economic assets and improve the resilience of the region's population.

Study area

Situated on the Gulf of Guinea in West Africa, Nigeria is administratively structured into six geopolitical zones. A map of the country depicting all six zones, including the North-East, North-West, North-Central, South-West, and South-East, is provided in Fig. 1a, while Fig. 1b shows the area of interest in this study. The South-South

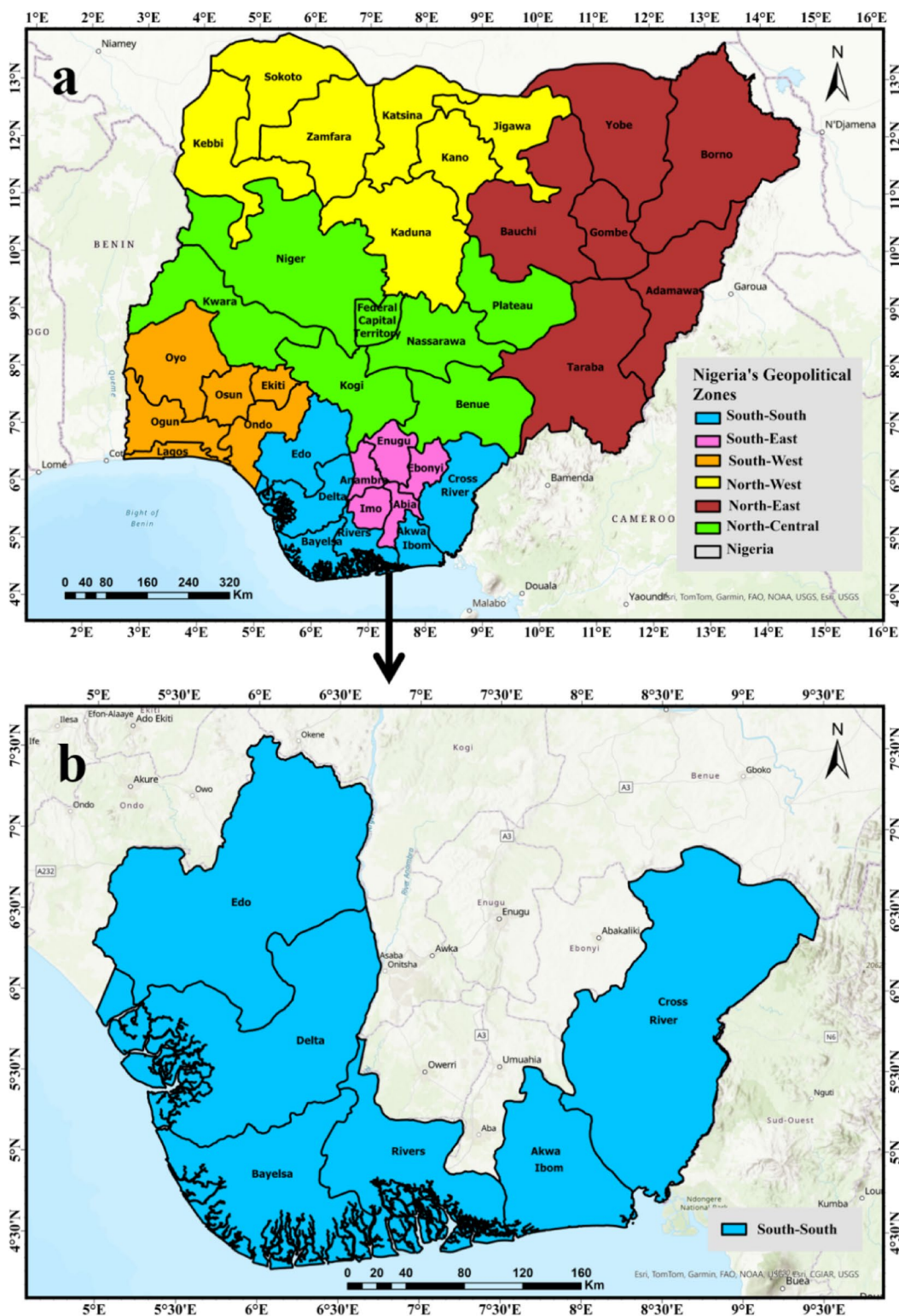


Fig. 1. (a) Map of Nigeria showing the 6 geopolitical zones and the encompassing states; (b) The map of the South-South geopolitical zone explicitly includes all 6 states that constitute the core of the Niger Delta region (study area).

geopolitical zone explicitly includes all six states that constitute the core of the Niger Delta, which represents the study area. These six states include: Akwa Ibom, Bayelsa, Cross River, Delta, Edo, and Rivers. Its approximate geographical extent is defined by latitudes 4°N to 8°N and longitudes 5°E to 9°E. The Niger Delta region, being largely coastal and sedimentary, includes significant Cretaceous, Tertiary, and more recent sediment deposits, as well as basement Precambrian units⁵⁰. The climate of the area can be classified as humid because it is located within the humid tropics. The region experiences average yearly temperatures between 22 and 29 °C⁵¹. The key climate variable is rainfall, which shows spatial variation across the study area. The region's climate is defined by two distinct seasons: a dry season from November to February and a wet season that spans from March to October. During the rainy season, rainfall can be very high, exceeding 3500 mm per annum, resulting in significant flooding that devastates many communities⁵².

Materials and methods

This study utilised the FR method to delineate flood-prone zones and produce a susceptibility map for the region. The methodological workflow, illustrated in Fig. 2, commenced with the compilation of a flood inventory and the selection of relevant conditioning factors (Table 1). The FR approach was subsequently used to quantify the spatial correlation between past flood events and these factors. From this analysis, the Relative Frequency (RF) for each factor class was determined and normalised to calculate a PR, or weight, for every factor. Subsequently, the final flood susceptibility map was produced, and the predictive performance of the model was ultimately validated by calculating the AUC.

Flood inventory map (FIM)

Flood susceptibility modeling is fundamentally reliant on a flood inventory, a dataset that maps the spatial history of flood events⁵³. This study focused on the severe flood of 2022 to create a flood inventory through multiple sources, including news reports, before and after satellite imagery analysis (<https://earthexplorer.usgs.gov/>), the 2022 United Nations Office for the Coordination of Humanitarian Affairs (UNOCHA) flood maps, NASA Earth Observatory reports, Google Earth satellite data, and detailed field surveys. The documented 142 flood locations were spatially referenced and analysed using GIS, with a standardised 70–30 percent split between training and

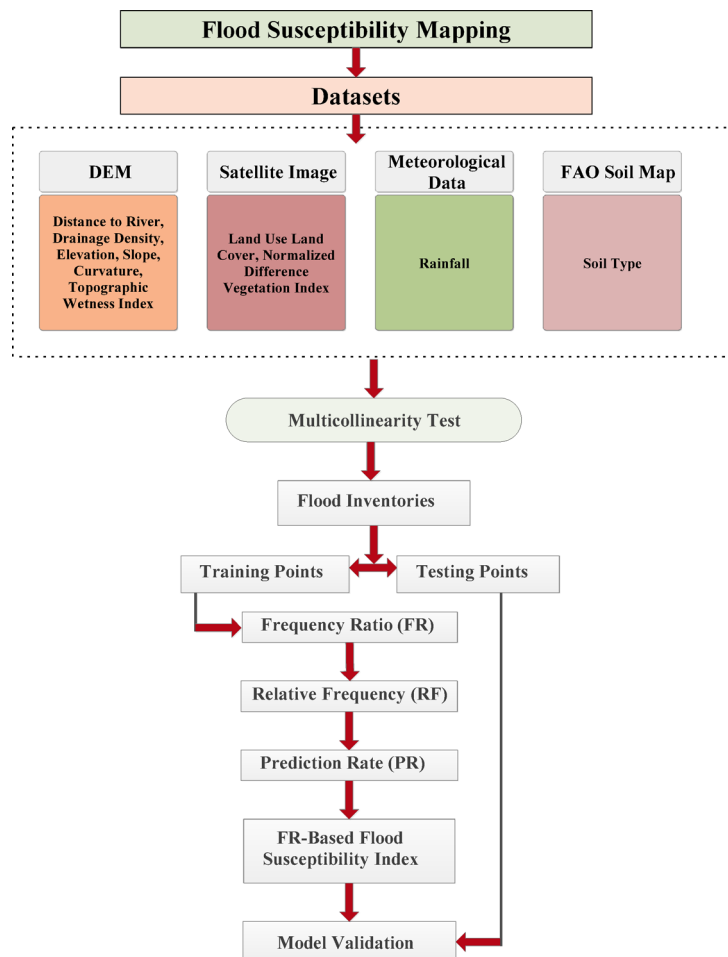


Fig. 2. Methodological sequence employed to develop the flood susceptibility map via the FR model technique.

Data	Data source	Year	Resolution/Scale/Layer
Rainfall (RF)	Crudata (https://crudata.uea.ac.uk/cru/data/hrg/)	2011–2020	0.5°0.5 Gridded
Distance to river (DR)	Calculated from DEM data (GIS)	–	–
Drainage density (DD)	Calculated from DEM data (GIS)	–	–
Land use land cover (LULC)	Sentinel 2 (https://dataspace.copernicus.eu/explore-data/data-collection/sentinel-data/sentinel-2)	2023	10 m
Elevation (EL)	USGS Earth Explorer (https://earthexplorer.usgs.gov)	2023	
Slope (SL)	Calculated from EL data (GIS)	–	–
Normalised Difference Vegetation Index (NDVI)	Sentinel 2 (https://dataspace.copernicus.eu/explore-data/data-collection/sentinel-data/sentinel-2)	2023	10 m
Soil type (ST)	https://www.fao.org/soils		1:5 000 000
Curvature (CU)	Calculated from DEM data (GIS)	–	–
Topography wetness index (TWI)	Calculated from DEM data (GIS)	–	–

Table 1. Summary of datasets used in the study.

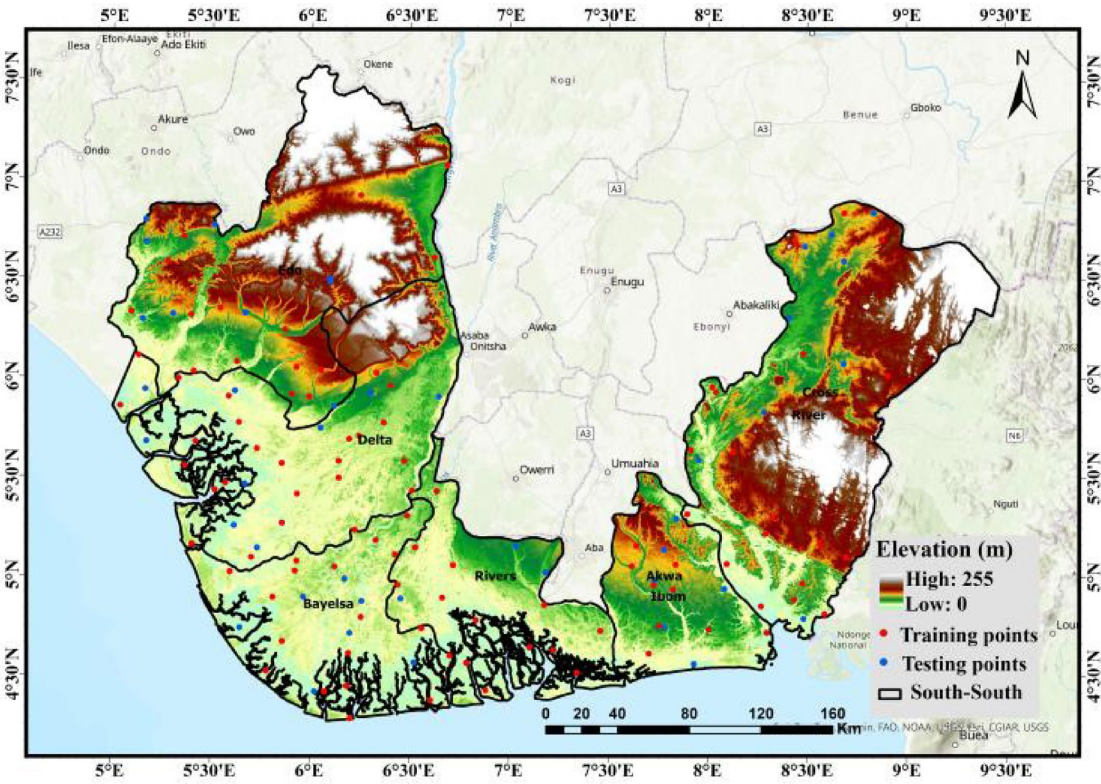


Fig. 3. Documented flood event locations in the study area.

testing datasets; a conventional approach in flood hazard modelling^{54–56}. The 70–30 split ratio for training and validation datasets is a commonly employed practice in flood susceptibility modelling and other geo-hazard assessments. Many empirical studies^{57–60} have adopted this ratio, particularly when utilising the FR model for flood susceptibility assessment, and this split ratio helps ensure robust model training and reliable performance assessment. The locations of past flood events are mapped in Fig. 3.

Flood influencing factors (FIFs)

The choice of conditioning factors in FR-based flood susceptibility studies often varies significantly across different research contexts, as no universal protocol exists. In this research, ten flood-influencing factors (FIFs) were chosen after ensuring the absence of multicollinearity. This selection was guided by a review of prior literature, data accessibility, expert consultation, and the specific environmental characteristics of the region. Using ArcGIS Pro, raster data layers were created for these ten parameters: Rainfall (RF), Distance to River (DR), Drainage Density (DD), Land Use/Land Cover (LULC), Elevation (EL), Slope (SL), NDVI, Soil Type (ST), Curvature (CU), and Topographic Wetness Index (TWI). As a key meteorological input, rainfall is a

direct precursor to flooding, where its intensity and frequency are critical determinants of flood events^{61,62}. The proneness of the study area to intense rainfall makes it particularly vulnerable to flooding¹⁰.

To analyse key hydrological factors, several datasets were processed. The average annual rainfall from 2011 to 2020 was obtained from the Climatic Research Unit (CRU) dataset (<https://crudata.uea.ac.uk/cru/data/hrg/>). The Inverse Distance Weighted (IDW) algorithm in ArcGIS Pro was employed to generate a seamless rainfall distribution map across the study area through spatial interpolation. The likelihood of flooding is strongly influenced by a location's proximity to river networks, as elevated water levels during heavy precipitation can cause rivers to overflow into adjacent floodplains. A raster dataset depicting distance to rivers was created by processing an SRTM DEM with the Euclidean Distance function in ArcGIS. This raster was subsequently classified with the Natural Breaks (Jenks) classification scheme. Drainage density was also incorporated, as a higher density facilitates more efficient surface runoff, thereby influencing flood susceptibility⁶³. Using the Line Density tool in ArcGIS, a drainage density map was generated based on a drainage network previously delineated from a flow accumulation raster.

Anthropogenic activities significantly modify LULC and the associated ecosystem services it provides⁶⁴. These continual LULC changes persistently alter hydrological processes, which can subsequently trigger flooding⁶⁵. To analyse these landscape characteristics in detail for the flood susceptibility assessment, a 10-m resolution LULC dataset for the year 2023 was sourced from the Sentinel-2 satellite via the Copernicus Data Space Ecosystem (<https://dataspace.copernicus.eu/explore-data/data-collections/sentinel-data/sentinel-2>).

To maintain uniformity for spatial analysis, all datasets were resampled to a consistent 30-m resolution by applying the nearest-neighbour interpolation technique in ArcGIS. Elevation is a key controlling factor in flood risk assessment^{66,67}, as it critically governs surface runoff patterns, flow direction, and inundation depth⁶⁸. This study utilised a 30-m resolution DEM obtained from SRTM through the USGS Earth Explorer platform (<https://earthexplorer.usgs.gov>). This DEM was subsequently processed in ArcGIS by executing a fill operation to remove all sinks and create a hydrologically corrected, depression less elevation model. Using the Natural Breaks (Jenks) method, the derived DEM was segmented into five distinct elevation classes. Slope gradient, which refers to the steepness of a terrain surface⁶⁹, is a critical parameter in flood risk assessment as it governs surface runoff velocity and direction. For this study, a slope map was derived from the DEM using the Slope tool in ArcGIS's Spatial Analyst module. The resulting slope raster was then categorised into five classes. The NDVI is a dimensionless indicator that quantifies vegetation density by analysing the contrast between near-infrared and visible light reflectance⁶⁹. Its values, which range from -1 to 1, are derived from a ratio-based calculation. For this research, an NDVI map was developed using a Sentinel-2A satellite image. The index was calculated using the Raster Calculator tool in ArcGIS with the following formula.

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)} = \frac{(Band\ 8 - Band\ 4)}{(Band\ 8 + Band\ 4)} \quad (1)$$

In this formula, the NDVI is derived from surface reflectance in the near-infrared (NIR, Band 8 for Sentinel-2A) and red (Band 4) spectral bands. Higher index values correspond to dense, healthy vegetation, while lower values typically indicate sparse vegetation or non-vegetated surfaces. The soil type map was developed using data from the FAO/UNESCO Soil Map of the World (<https://www.fao.org/soils-portal>), which was extracted and converted to raster format in ArcGIS. To ensure uniformity for analysis, all raster factor maps were standardised. Each factor map was reclassified onto a standardised ordinal scale from 1 (very low susceptibility) to 5 (very high susceptibility) through the Reclassify function in ArcGIS. Additionally, curvature, a topographic parameter affecting surface runoff, was derived from the DEM via the curvature tool in ArcGIS's Spatial Analyst module. Terrain curvature, classified as concave, convex, or flat, controls runoff dynamics and acts as a key indicator of flood-prone areas. The TWI is a commonly used metric for assessing the influence of topography on hydrology and for predicting spatial patterns of soil moisture⁷⁰. This index quantifies the propensity for water to converge at a given location and its likelihood to be transported downslope by gravitational forces. As noted by Lappas and Kallioras⁷¹, elevated TWI values are associated with drainage depressions, low-lying, flat areas with saturated soils, while lower values correspond to ridges and crests characterised by steeper slopes. Consequently, areas with elevated TWI values demonstrate a greater propensity for flooding. The computation of the TWI for this analysis was performed with the Raster Calculator function within the ArcGIS Spatial Analyst toolbox, as defined by the following formula:

$$TWI = \ln \left(\frac{A_s}{\tan \beta} \right) \quad (2)$$

In the given formula, the variable 'As' refers to the specific catchment area (m² m⁻¹), and 'β' represents the slope angle expressed in degrees.

Multicollinearity diagnosis method for flood susceptibility factors

When two or more independent variables in a statistical model are linearly related, it creates a multicollinearity problem. This correlation often leads to unreliable and imprecise estimates of the model's coefficients, making the results difficult to interpret⁷². The assessment of multicollinearity, a critical step preceding model implementation in flood studies, can be conducted using multiple established methods⁷³. In this study, Tolerance and Variance Inflation Factor (VIF) values were derived for all influencing factors to evaluate multicollinearity. Following Khosravi, et al.⁷⁴, a VIF value exceeding 9 or a Tolerance value falling below 0.1 indicated a multicollinearity problem. In line with Mitra and Das⁷⁵, any factor meeting these criteria was removed before model development. These metrics were derived for each variable through Eqs. (3) and (4).

$$T_i = 1 - R_i^2 \quad (3)$$

$$VIF_i = \frac{1}{T_i} \quad (4)$$

Here, T_i is the tolerance, VIF_i is the variance inflation factor associated with the i^{th} predictor variable, and R_i^2 is the coefficient of determination from the corresponding regression model.

Frequency ratio method

In flood susceptibility modeling, the FR is a prevalent bivariate method. It functions by analysing the spatial relationship between a dataset of past flood events (the dependent variable) and the independent conditioning factors that contribute to flooding⁷⁶. Yilmaz⁷⁷ describes FR as a fundamental statistical analysis that evaluates the specific impact of each criterion on flood prediction. The method's continued relevance is driven by several factors, one of which is its user-friendliness, a direct result of its uncomplicated and transparent logic⁷⁸. The FR method features an intuitive workflow, with clear input, calculation, and output stages that are easily implemented within a GIS framework. This operational simplicity contributes to its status as a straightforward and commonly adopted technique for flood risk evaluation. A further advantage is its dual capacity to produce flood risk maps and to facilitate the analysis of relationships between flood risk and its contributing factors. For each category of the conditioning factors, the ratio of flood-affected area to the total area was computed using training data derived from the flood inventory. This calculation, following Addis³², equivalently, represents the probability of flood occurrence relative to non-occurrence for that specific class. An FR value exceeding 1.0 signifies a strong, positive correlation with flooding and indicates a high probability, whereas a value below 1.0 denotes a negative correlation and lower susceptibility. These values were derived using Eq. 5.

$$FR = \left[\frac{N_{pix}(SX_i)}{\sum_{i=1}^m SX_i} \right] / \left[\frac{N_{pix}(X_j)}{\sum_{j=1}^n N_{pix}(X_j)} \right] \quad (5)$$

In this equation, $N_{pix}(SX_i)$ represents the number of flooded pixels in class i of factor X , while $N_{pix}(X_j)$ denotes the total number of pixels comprising the entire factor X_j . The parameters m and n correspond to the number of classes in factor X_i and the total quantity of factors examined in the research, respectively. The resulting FR was then converted to a probability value between 0 and 1 through normalisation using Eq. 6, thereby obtaining the RF.

$$RF = \frac{FR_{ij}}{\sum FR_j} \quad (6)$$

In this context, RF signifies the Relative Frequency for a specific factor class, calculated from the Frequency Ratio (FR_{ij}) of class i in factor j divided by the sum of all FR values ($\sum FR_j$) for that factor. A constraint of the RF method is that it assigns equal importance to all factors after normalisation. To overcome this limitation, the PR was subsequently computed using Eq. 7 to quantify the relationship between each conditioning factor and the flood inventory data.

$$PR = \frac{RF_{maxj} - RF_{minj}}{\text{Min}(RF_{maxj} - RF_{minj})} \quad (7)$$

The PR for each factor is defined by this equation, which utilises the factor's maximum (RF_{maxj}) and minimum (RF_{minj}) RF values. In accordance with Ali, et al.⁷⁸, the calculated RF values were assigned to their corresponding reclassified factor classes. The Flood Susceptibility Index (FSI) was subsequently produced in ArcMap by executing Eq. 8. The final FSI output was classified into five separate susceptibility categories using the natural breaks method.

$$FSI_{FR} = \sum_{i=1}^n PR_i \times RF_i \quad (8)$$

Here, FSI_{FR} is the FR-based flood susceptibility index, computed from the Prediction Rate (PR_i) and relative frequency (RF_i) of each factor class across all n factors.

Model validation

The validation of a model is an essential process for confirming its predictive accuracy and dependability, a principle that applies directly to flood risk models⁷⁹. A significant oversight in flood risk research is the frequent omission of model validation, a step essential for establishing predictive credibility⁸⁰. Nigerian case studies exemplify this trend, with numerous assessments failing to incorporate validation protocols^{81–83}. This study bridges this crucial gap through ROC-AUC validation, following established procedures in recent studies^{49,84}. Several validation approaches exist for evaluating predictive models. This study employed the ROC-AUC method, a robust statistical technique widely recognised for assessing natural hazard models, specifically for flood mapping applications, owing to its logical framework and straightforward analytical interpretation^{38,85}. The validation process involved comparing all 142 flood inventory points with the flood susceptibility index through

ROC-AUC analysis. This technique evaluates a model's predictive performance by plotting its sensitivity against its false positive rate (1-specificity) across all classification thresholds⁸⁶. The sensitivity and specificity were computed using Eq. 9 and 10. For generating and analysing the ROC curve, this study employed the specialised ROCR package available in the R programming platform.

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (9)$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (10)$$

Here, TP denotes flood points accurately classified as flood class, TN indicates non-flood points correctly assigned to non-flood class, FN refers to flood points erroneously categorised as non-flood class, and FP signifies non-flood points incorrectly classified as flood class.

Results

Flood susceptibility indicators

Through comprehensive spatial analysis of the flood influencing factors, the regional flood susceptibility patterns were systematically evaluated and visualised as indicated in Fig. 4 and subsequently reclassified to a scale from 1 (very low susceptibility) to 5 (very high susceptibility) as shown in Supplementary Fig. S1.

Multicollinearity evaluation

To evaluate potential multicollinearity among the FIFs, random sample points across the study area were used. The multicollinearity analysis incorporated 300 sample points to evaluate the Variance VIF and tolerance metrics of FIFs. The statistics inferred that none of the chosen FIFs has collinearity issues (Table 2). The established diagnostic thresholds for multicollinearity are $\text{VIF} > 9$ and $\text{tolerance} < 0.1$.

FR model

To quantify the contribution of different flood-influencing factors, the FR was computed for each class relative to known flood inventory points. The analysis involved reclassifying all factors into five classes and calculating their respective FR values using training points. In this context, a higher FR value directly reflects a greater probability of flood occurrence for that specific factor class. According to the FR results, the FR values for the rainfall class range from 0.17 to 1.56. The lowest range (92.061–639.495) has a very low FR of 0.17, indicating low susceptibility. The analysis reveals distinct patterns between flood occurrence and various conditioning factors. For rainfall, a clear positive correlation exists, where higher precipitation amounts correspond to increased flood probability. The highest rainfall class (2139.467–2883.978) shows the most significant correlation with an FR of 1.56. Proximity to rivers is another critical factor; the FR value is highest (1.15) for areas closest to the river (0–2070.23 m) and systematically decreases with increasing distance. Similarly, drainage density shows a direct relationship with flooding, as higher density values lead to higher FRs, peaking at 1.18 for the class of 0.239–0.413 km/km². LULC analysis indicates that Waterbodies/Flooded Vegetation and Built-Up/Bare Surfaces are most associated with flooding. In contrast, Crop Land, Rangeland, and Forest areas demonstrate a markedly lower flood likelihood, suggesting a protective effect. Topographic factors also play a major role. Lower elevations are substantially more susceptible, with FR values of 1.48 and 1.27 for the 0–39 m and 39–88 m classes, respectively, while higher elevations have values below 0.23. An inverse relationship is evident for slope, where gentler slopes (0–0.918) have a higher FR (0.89) than steeper terrain. The NDVI results show that areas with sparse vegetation (NDVI: –1 to 0.043) have an elevated FR of 1.16, indicating higher runoff and flood susceptibility. Conversely, regions with dense, healthy vegetation (NDVI: 0.427 to 1) exhibit a much lower FR of 0.28, highlighting the role of vegetation in promoting water absorption and reducing flood risk. Analysis of soil types indicates that clay loam soil exhibits a stronger correlation with flooding, demonstrated by an FR of 1.22. In terms of topography, flat surfaces (curvature class –0.0004 to 0.0003) are highly susceptible to flooding, reflected in an FR value of 1.12. The TWI also shows a significant relationship, with the class 11.157–23.270 having a high FR of 1.20. The FR for each factor class was derived by normalising its FR value to a scale of 0 to 1. The PR for each factor, which quantifies its overall importance, is summarised in Table 3. The results identify elevation as the most influential factor with a PR of 15.98. It is followed by rainfall (13.28), soil type (11.05), drainage density (9.97), NDVI (8.48), LULC (7.16), distance to river (7.0), slope (6.89), TWI (1.67), and curvature (1.00). A greater PR value indicates that the factor plays a more substantial role in controlling flood susceptibility within the study region. The ultimate FSI was produced by aggregating the products of each factor class's FR and its associated factor-level PR.

Delineation of flood-prone areas

The application of the Jenks natural breaks algorithm to the FR model output produced a flood susceptibility map with five discrete categories, ranging from very low to very high risk (Fig. 5). Spatially, the analysis shows that the high and very high susceptibility categories are the most dominant, encompassing 21,721.36 km² (59.295%) and 4323.93 km² (11.803%) of the study area, respectively. Conversely, the low susceptibility category constitutes 28.212% (10,334.74 km²) of the region. The moderate and very low susceptibility zones represent significantly smaller proportions, at just 0.688% (251.92 km²) and a minimal 0.002% (0.72 km²), respectively (Table 4). The generated map shows that the coastal and southern sections of Bayelsa, Rivers, and Delta states are predominantly characterised by very high flood susceptibility, depicted in red, signifying their extreme vulnerability to flooding. Specifically, within Bayelsa, the highly susceptible areas encompass Ekeremor, Southern Ijaw, and Brass. In Rivers State, this high susceptibility extends to Port Harcourt, Okrika, Degema, Khana, and Asari-Toru. For Delta State,

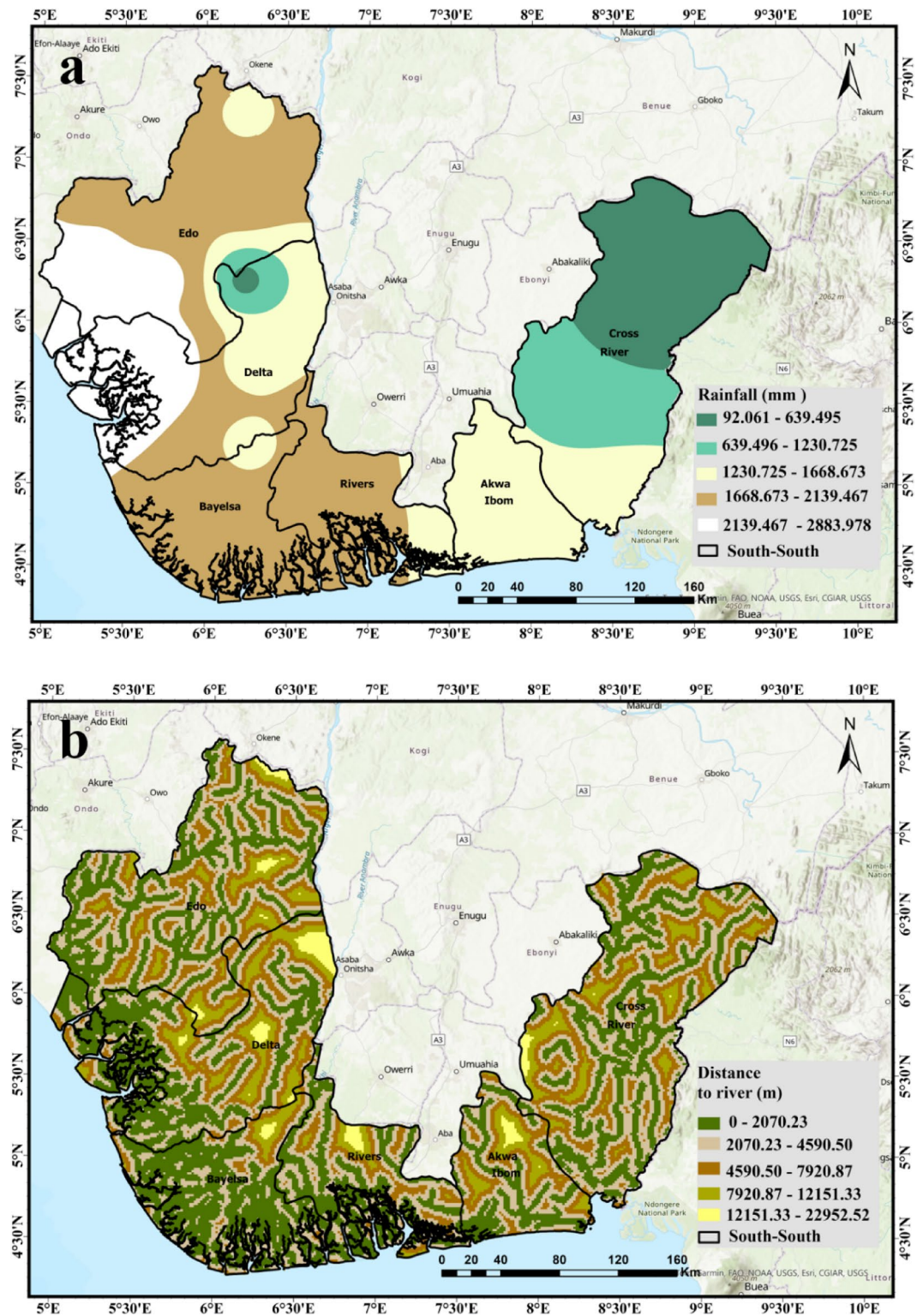


Fig. 4. Flood conditioning factors (A) Rainfall (RF) (B) Distance to river (DR) (C) Drainage density (DD) (D) Land use land cover (LULC) (E) Elevation (EL) (F) Slope (SL) (G) Normalized Difference Vegetation Index (NDVI) (H) Soil type (ST) (I) Curvature (Cu) (J) Topographic wetness index (TWI).

the areas of concern include Warri North, Warri South and Burutu. This is expected as it is a low-lying delta region characterised by numerous rivers and creeks, making it extremely vulnerable to flooding from heavy rainfall, river overflows, and storm surges from the Atlantic Ocean. Flood susceptibility is predominantly high across a substantial part of the region. The moderate zone is a transition area which can be found mainly in Edo and Cross River. Similarly, the low flood susceptibility zones are dominant in Edo, Cross River and the northern

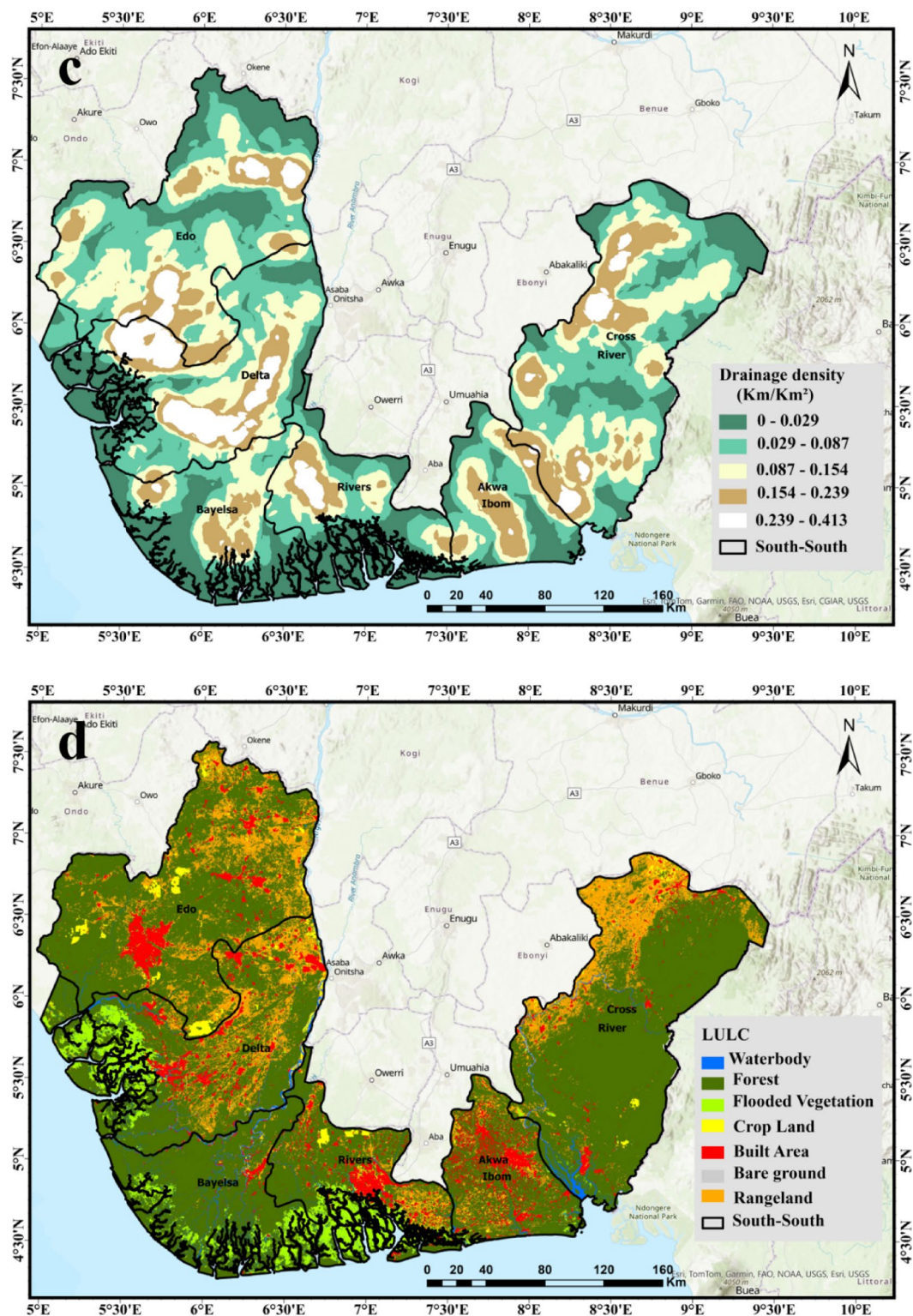


Fig. 4. (continued)

part of the Delta. In Edo, areas like Owan East, Esan West, Esan Central, and Igueben fall under this category. In Cross River, areas such as Obanliku, Obudu, Yakurr and Akamkpa fall under this category. In Delta, areas in the northern region, such as Ika South, Ika North East and Aniocha South fall under this category. Areas of very low susceptibility, while limited in extent, are sporadically distributed in locations such as Akamkpa in Cross River and Ika North East in Delta State. Overall, the Niger Delta region is predominantly classified as having high to very high flood susceptibility. This highlights the necessity of implementing targeted mitigation strategies to reduce potential flood impacts in the most vulnerable areas.

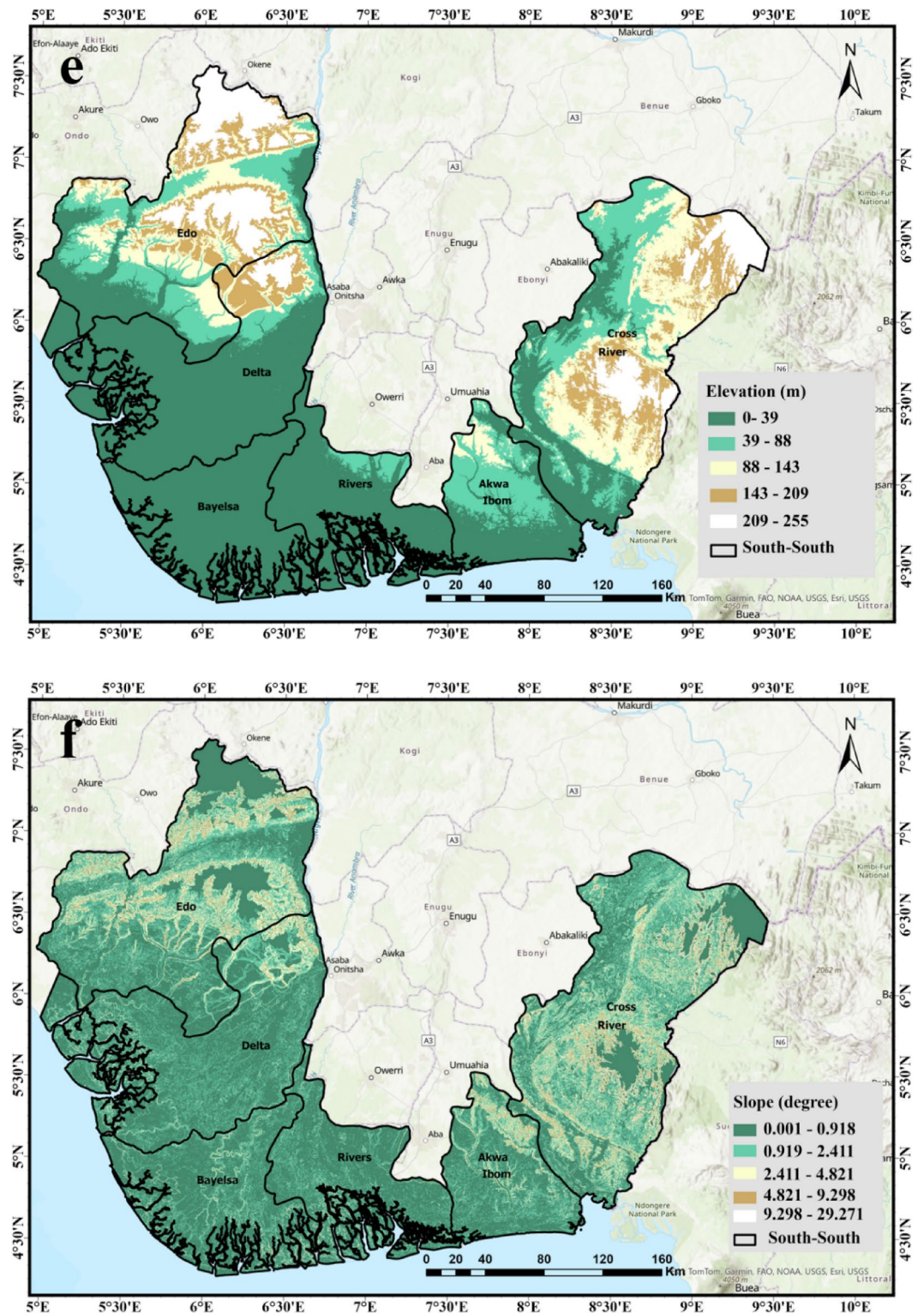


Fig. 4. (continued)

Model performance evaluation

The reliability of the flood susceptibility map produced by the FR model must be verified through validation to ensure its practical utility. To evaluate the predictive performance of the model in this research, the AUC of the ROC was utilised. This metric is a conventional measure of diagnostic accuracy, derived from a plot of the true positive rate against the false positive rate. The AUC value ranges from 0.5, indicating a performance no better than random chance, to 1.0, representing a perfect prediction, and it is this value that determines the model's

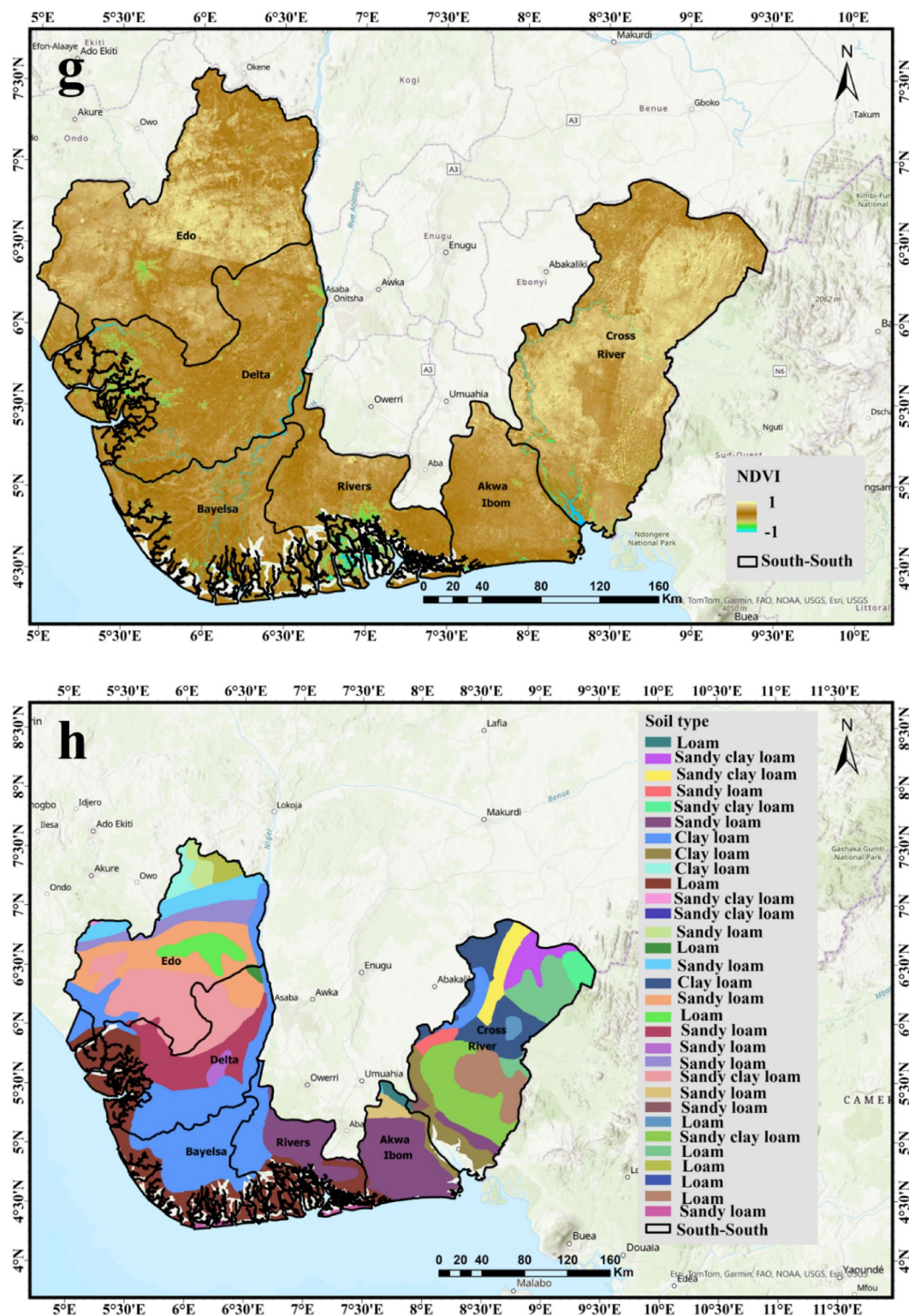


Fig. 4. (continued)

classification accuracy. According to the classification by Yesilnacar and Topal¹⁸⁷, an AUC of 0.9–1.0 is ‘excellent’, 0.8–0.9 is ‘very good’, 0.7–0.8 is ‘good’, 0.6–0.7 is ‘average’, and 0.5–0.6 is ‘unsatisfactory’.

The validation results demonstrated an AUC of 0.8316 for the FR model (Fig. 6), indicating a commendable level of forecasting accuracy. This robust performance confirms that the FR model is a capable and appropriate methodology for generating flood susceptibility maps in the context of this study, a finding consistent with previous literature^{32,41,88}.

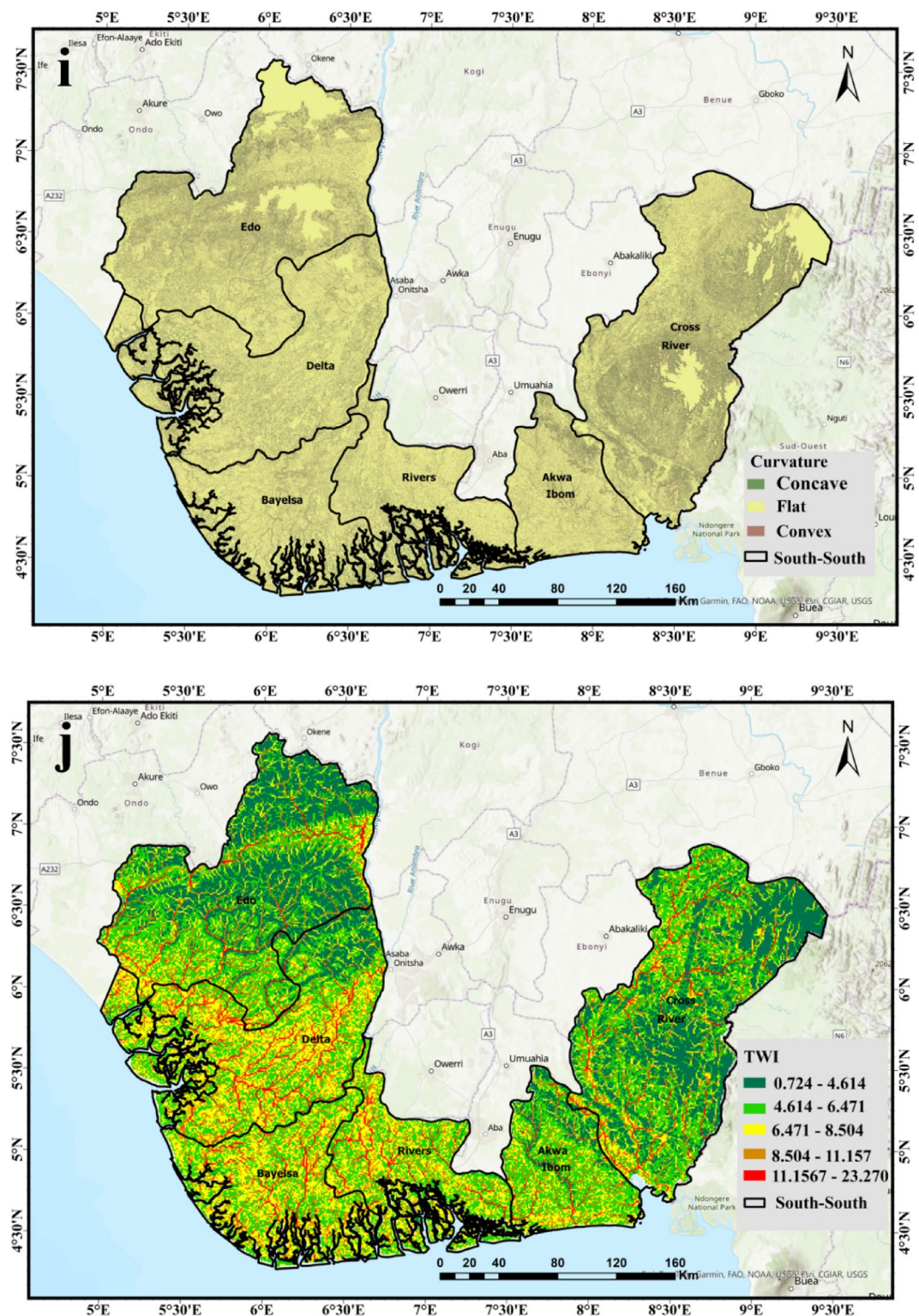


Fig. 4. (continued)

Parameters	RF	DR	DD	LULC	EL	SL	NDVI	ST	CU	TWI
Tolerance	0.93	0.87	0.98	0.87	0.73	0.44	0.93	0.84	0.41	0.94
VIF	1.07	1.15	1.02	1.15	1.37	2.29	1.07	1.19	2.45	1.06

Table 2. Multicollinearity statistics (Tolerance and VIF) of FIFs.

FIF	Class	Area (Km ²)	% Area	Flood pixels	% pixels	FR	RF	PR
RF	92.061–639.495	10,985.7942	13.3	11,266	1.78	0.17	0.06	13.28
	639.495–1230.725	8407.8288	10.2	5677	0.89	0.34	0.11	
	1230.725–1668.673	19,688.9112	23.8	121,709	19.19	0.61	0.19	
	1668.673–2139.467	31,827.3039	38.5	60,307	9.51	1.25	0.21	
	2139.467–2883.978	11,669.8743	14.1	435,191	68.63	1.56	0.63	
DR	0–2070.23	33,186.7917	40.2	12,522	1.98	1.15	0.54	7.00
	2070.23–4590.50	25,918.506	31.4	12,819	2.03	0.57	0.31	
	4590.50–7920.87	15,181.4394	18.4	39,088	6.19	0.23	0.12	
	7920.87–12,151.33	6849.3303	8.3	127,711	20.24	0.05	0.03	
	12,151.33–22,952.52	1463.157	1.8	438,794	69.55	0.03	0.01	
DD	0–0.029	21,087.225	25.5	45,617	7.19	0.08	0.05	9.97
	0.029–0.087	23,925.6927	29.0	77,392	12.20	0.13	0.31	
	0.087–0.154	20,741.2326	25.1	95,844	15.11	0.76	0.5	
	0.154–0.239	11,995.4025	14.5	165,180	26.05	0.98	0.08	
	0.239–0.413	4821.6771	5.8	250,167	39.44	1.18	0.05	
LULC	WB/Flooded Vegetation	6280.4016	7.6	37,256	9.75	1.22	0.09	7.16
	BA/Bare Surfaces	5754.5244	7.0	62,683	16.40	1.07	0.05	
	Crop Land	1496.7387	1.8	226,086	59.17	0.52	0.43	
	Rangeland	11,942.4195	14.5	29,730	7.78	0.19	0.36	
	Forest	57,078.9468	69.1	26,311	6.89	0.04	0.08	
EL	0–39	43,025.3469	52.1	622,648	98.17	1.48	0.99	15.98
	39–88	13,527.9657	16.4	5833	0.92	1.27	0.62	
	88–143	11,039.8968	13.4	5495	0.87	0.23	0.23	
	143–209	7280.1495	8.8	183	0.03	0.11	0.18	
	209–255	7701.0651	9.3	100	0.016	0.07	0.05	
SL	0–0.918	44,707.7718	54.6	414,626	67.03	0.89	0.08	6.89
	0.918–2.411	24,346.3104	29.7	163,710	26.46	0.76	0.44	
	2.411–4.821	9640.8234	11.8	37,585	6.08	0.59	0.37	
	4.821–9.298	2600.5743	3.2	2607	0.42	0.24	0.04	
	9.298–29.271	569.817	0.7	81	0.013	0.18	0.08	
NDVI	–1–0.043	1606.4091	2.0	133,142	21.03	1.16	0.22	8.48
	0.043–0.215	8012.8116	10.0	154,048	24.34	0.69	0.16	
	0.215–0.333	20,250.027	25.4	155,757	24.61	0.55	0.22	
	0.333–0.427	30,884.787	38.7	123,533	19.52	0.32	0.19	
	0.427–1	19,046.7369	23.9	66,522	10.51	0.28	0.21	
ST	Sandy	10,189.9449	12.5	59,472	10.82	0.54	0.13	11.05
	Sandy loam	28,662.4233	35.3	1914	0.35	0.81	0.2	
	Loam	30,086.1153	37.0	324,377	58.99	0.74	0.18	
	Sandy clay	8876.3319	10.9	159,718	29.05	0.93	0.23	
	Clay loam	3452.2245	4.2	4418	0.80	1.22	0.25	
CU	–0.0081––0.0004 (Concave)	11,019.5811	13.4	110,007	24.29	0.57	0.34	1
	0.0003–0.0078 (Convex)	15,536.7927	18.9	372,182	5.87	0.09	0.06	
	–0.0004–0.0003 (Flat)	55,457.1108	67.6	108,315	69.84	1.12	0.6	
TWI	0.724–4.614	17,280.0675	22.0	51,593	16.92	0.21	0.21	1.67
	4.614–6.471	28,743.3774	36.6	97,553	31.99	0.36	0.18	
	6.471–8.504	20,215.3203	25.7	89,728	29.42	0.45	0.19	
	8.504–11.157	9050.0202	11.5	51,856	17.00	0.68	0.2	
	11.157–23.270	3335.1894	4.2	14,225	4.66	1.20	0.23	

Table 3. Frequency ratio (FR), Relative frequency (RF), and Prediction rate (PR) Values for FIF classes.

Discussion

The study employs the FR model, a statistical technique widely used in predictive modelling to evaluate and spatially delineate flood-prone areas^{18,33,59}. The use of geospatial techniques, such as ArcGIS software for processing datasets, aligns with standard practices in flood susceptibility mapping^{22,89}. The analysis incorporated ten conditioning factors: Rainfall, Distance to River, Drainage Density, LULC, Elevation, Slope, NDVI, Soil Type, Curvature, and TWI, which are consistent with those considered in other flood susceptibility assessments

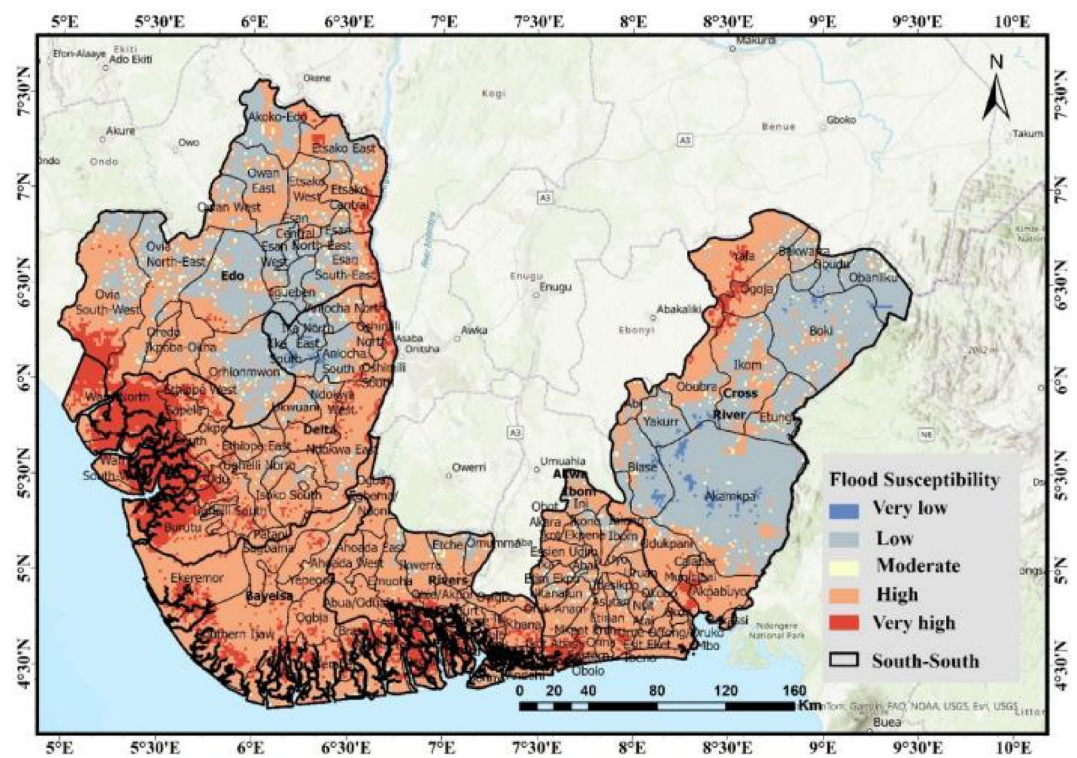


Fig. 5. Geographic distribution of flood-prone areas in the Niger Delta Region of Nigeria.

Flood susceptibility	Area (Km ²)	Area (%)	Class ratings
Very low	0.72	0.002	1
Low	10,334.736	28.212	2
Moderate	251.9181	0.688	3
High	21,721.3596	59.295	4
Very high	4323.9339	11.803	5

Table 4. Flood susceptibility coverage.

in various regions^{18,90}. The resulting flood susceptibility map identifies the coastal and southern regions of Bayelsa, Rivers, and Delta states as the zones of highest risk. This zone covers 4323.9339 km² (11.803%). High susceptibility zones are the most extensive, occupying an area of approximately 21,721.36 km² (59.295%) of the study region. This elevated susceptibility is attributable to the region's coastal geography, which features low-lying topography and consequent high exposure to flooding⁹¹. This flat topography, coupled with numerous river distributaries, contributes significantly to its flood vulnerability⁹². Coastal areas are particularly susceptible to tidal movements and rising sea levels, which can cause flooding in the region¹⁰. This outcome corroborates the findings reported by Peter, et al.⁹³, Chinedu, et al.²⁵, Ozegin and Ilugbo⁹⁴, and Horsfall, et al.⁹¹.

Consistent with its established role in flood generation, rainfall demonstrates a strong influence in the model. The FR confirms a positive correlation, where the highest precipitation class (2139.467–2883.978 mm) yields an FR of 1.56, compared to a value of 0.17 for the lowest class (92.061–639.495 mm). This pattern highlights rainfall as a primary driver of flooding in the Niger Delta region, a result that corresponds to the study of Echendu¹³ and the area's environmental and topographic context previously described in Section “Study area”. The Niger Delta region's pronounced flood susceptibility is heavily influenced by its physiographic and climatic context. As a low-lying coastal zone with high annual rainfall, it experiences substantial runoff, directly increasing its flood risk. This study's FR values for key factors such as distance to rivers, drainage density, elevation, slope, NDVI, and TWI align with established hydrological principles, consistently demonstrating a direct and strong correlation with flooding in the region. Furthermore, the topographic characteristic of curvature also conforms to expectations; flat surfaces (curvature class –0.0004 to 0.0003), which inhibit water dispersal and promote accumulation, yielded the highest FR value of 1.12, confirming their heightened vulnerability⁴². The analysis of land curvature further substantiates established geomorphological principles, with concave and convex forms showing FR values of 0.57 and 0.09, respectively, a trend supported by existing literature^{95,96}. In LULC, the highest FR value (1.22) is logically associated with Waterbody/Flooded Vegetation, as these zones are inherently prone to inundation⁹⁷. Similarly, Built-Up Areas and Bare Surfaces, with their impermeable surfaces, demonstrate

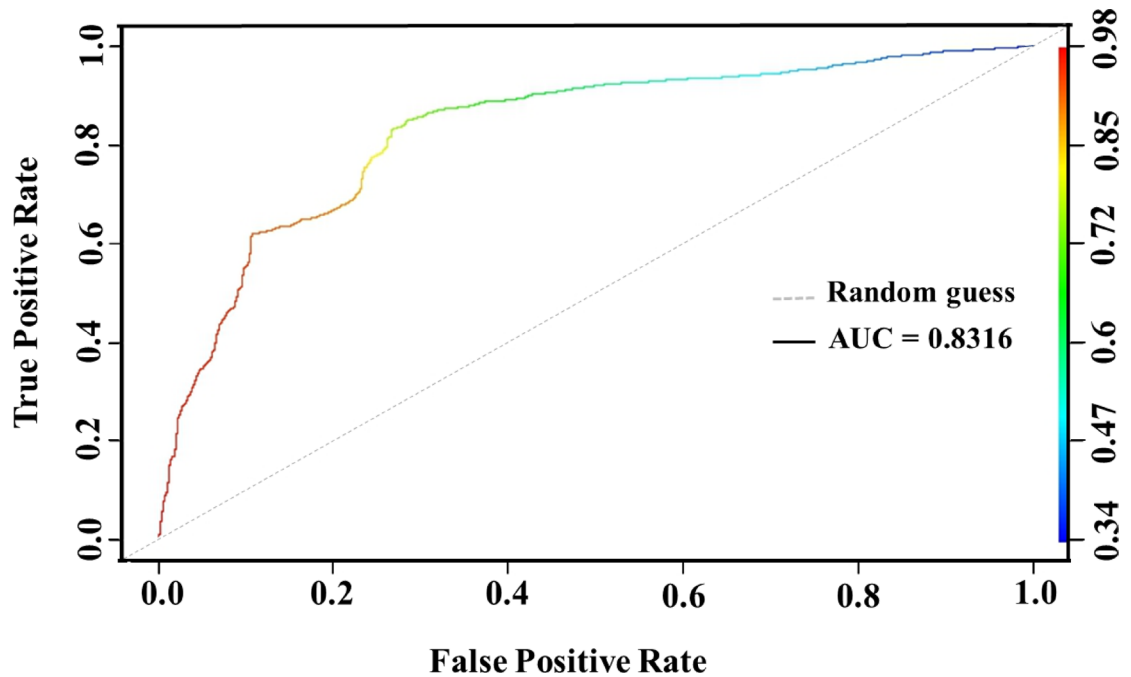


Fig. 6. Predictive accuracy assessment of the flood susceptibility map using AUC.

a strong positive correlation ($FR > 1$) by promoting surface runoff⁹⁸. Urbanisation exacerbates flood hazards through the creation of impervious surfaces that prevent water infiltration and channel rainfall directly into runoff^{99,100}. Crop land, rangeland, and forest all yielded FR values below 1.0, indicating a negative correlation with flooding. Forests demonstrated the most pronounced mitigating effect, with the lowest FR value of 0.04. This aligns with their well-documented role in flood risk reduction, as they enhance soil water retention and delay the concentration of runoff, leading to their consistent classification as very low-risk areas^{101,102}. The soil type analysis reveals a clear susceptibility pattern. Clay loam, with an FR of 1.22, demonstrates a high correlation with flooding, which is consistent with its high content of fine silt and clay particles that limit infiltration. This is followed by sandy clay, sandy loam, and loam. In contrast, sandy soil yielded a low FR of 0.54, reflecting its coarse-grained structure that promotes rapid drainage and thus reduces flood propensity.

The Niger Delta is a sedimentary basin primarily composed of unconsolidated sands, silts, and clays deposited by the Niger River and its tributaries¹⁰³. This geomorphology, characterised by low-lying coastal plains and numerous river distributaries, inherently contributes to its flood vulnerability¹⁰⁴. Shoreline changes along the Niger Delta coast have been linked to shifts in hydroclimate and human activities over the last century¹⁰⁵. The arcuate Niger Delta shoreline experiences continuous morphological changes, with erosion often outweighing accretion¹⁰⁶. Accretion increased by 2.91% and erosion by 4.25% over 30 years from 1986 to 2016¹⁰⁶. Major erosion cases are often due to a changing climate and the velocity of flow of major estuaries into the ocean. Human activities such as sand mining, the presence of hard and engineered structures on the shore, and other commercial and social activities also contribute to shoreline changes¹⁰⁶. Sandbars, which are active geomorphic features of rivers, provide important habitats, but their formation is impaired by altered flow and sediment regimes due to human development¹⁰⁷. Similarly, large-scale upstream impoundment in dams and reservoirs along the Niger River has drastically reduced sediment delivery to the Niger Delta coast¹⁰⁸. This reduction in fluvial sediment inputs following dam construction is a significant factor in coastal erosion in West Africa, and this can significantly impact the stability of the delta's coastline and its susceptibility to erosion and flooding¹⁰⁹.

Climate change, characterised by global warming, increased frequency and intensity of precipitation, and extreme weather events, poses a significant threat to vulnerable regions like the Niger Delta. The Niger Delta, already susceptible to sea level rise, storm surges, coastal erosion, and flooding, faces exacerbated vulnerability due to oil spillages, gas flaring, and environmental degradation¹¹⁰.

Recent advancements in ML are offering new avenues for flood susceptibility mapping and disaster management, particularly relevant for understanding climate change impacts in the Niger Delta. While ML models often demonstrate higher accuracy and generalizability in complex, data-rich environments¹¹¹ this study utilised the FR model due to its transparent interpretability and robust performance with limited historical flood inventory data, a common challenge in many regions in Africa. This transparency is invaluable for local stakeholders and policymakers who may require straightforward, evidence-based tools for initial flood risk assessment and land-use planning, especially where vast labelled datasets for complex ML algorithms remain inaccessible. However, it is important to acknowledge that recently, studies in some parts of Nigeria have begun to explore the application of ML algorithms such as Random Forest¹¹², Support Vector Machine (SVM)¹¹³, and Artificial Neural Networks (ANN)¹¹⁴ for flood risk assessment. This trend indicates that, as more comprehensive flood inventories and high-resolution spatial and temporal data become available, similar approaches can be

	Evidential belief function (EBF)	Weight of evidence (WoE)	Information value (IV)	Frequency ratio (FR)
Study	Ghosh ¹¹⁶	Mousavi et al. ¹¹⁷	Addis ³²	This study
AUC value	0.7219	0.7686	0.8000	0.8316
Data set of 30 m resolution	Yes	Yes	Yes	Yes

Table 5. Comparison of this study with other bivariate statistical models.

applied in other regions of the country, including the Niger Delta. Such advancements have the potential to provide a more robust framework for assessing flood patterns, identifying key environmental drivers, and forecasting future flood-prone areas under changing climatic conditions.

Limitations of the study and directions for future research

A primary limitation of this study, however, lies in the model’s data-dependent nature, as the precision of its outputs is contingent upon the completeness and accuracy of the underlying flood inventory data. This investigation sourced its flood event data mainly from previously established maps, which were quite limited, documented using point references and might not fully reflect the full extent of actual flooding. Furthermore, as a statistically based approach, the FR model identifies spatial correlations but does not simulate the physical processes of flooding, resulting in uncertainties, as it cannot predict flood depth or velocity. In addition, the reliance on a medium-resolution (30 m) open-source DEM for key geomorphological factors, while practical and widely adopted, inherently provides a generalised representation of terrain conditions, which could be further refined with higher-resolution data. To advance this research, several pathways are recommended. Future work should prioritise the use of high-resolution topographic data and more comprehensive flood inventories to enhance model accuracy. The FR model’s predictive capability could also be significantly improved by integrating it into hybrid frameworks that combine it with enhanced ML techniques as high-resolution spatial and temporal flood data becomes available. Finally, to move beyond susceptibility mapping and towards a mechanistic understanding of flood hazards, subsequent studies should couple these methodologies with hydraulic simulation models like HEC-RAS or HEC-HMS. Such an integrated approach would provide a more robust and dynamic analysis of flood risk in the region.

Despite these limitations, the FR-based methodology establishes a reproducible foundation for regional flood susceptibility mapping, readily adaptable to other data-constrained environments.

Practical implications and study contributions

The findings have significant implications for real-world applications, enabling policymakers to prioritise flood mitigation in high-susceptibility zones across the six core states (Akwa Ibom, Bayelsa, Cross River, Delta, Edo, and Rivers) of the Niger Delta, which produce over 90% of Nigeria’s oil¹¹⁵. The multicollinearity test helps to ensure that the FIFs selected are independent of each other, preventing redundancy and model instability while enhancing the reliability of spatial relationships quantified by the FR model. This study represents the first application of the FR model across the unified Niger Delta region, spanning its six core states. The FR model is compared with other existing bivariate statistical models in Table 5, highlighting its better performance. The FR model stands out for its exceptional simplicity and user-friendliness compared to Evidential Belief Function (EBF), Weight of Evidence (WoE), and Information Value (IV) in bivariate statistical flood susceptibility analysis. The result of this study, therefore, highlights the potential of FR to efficiently quantify the spatial correlation between flood occurrences and contributing factors, providing clear insights into the relative importance of each variable.

Conclusion

The Niger Delta region of Nigeria confronts a significant environmental challenge with widespread flooding, which endangers a vital national economic zone recognised for its substantial oil and gas production. Such flood occurrences can severely impede energy operations, devastate infrastructure, and detrimentally affect local communities, thereby obstructing sustainable socio-economic advancement. Therefore, robust flood management strategies, supported by scientifically derived hazard maps, are indispensable for mitigating potential damage.

An assessment of flood susceptibility was carried out utilising an FR model integrated within a Remote Sensing-GIS framework. Through the analysis of ten multicollinearity-free conditioning factors, the study determined that 71.098% of the region, covering an area of 26,045.29 km², falls into categories of high to very high flood susceptibility. This finding emphasises the critical need for focused mitigation efforts and informed land-use planning. The model’s strong predictive accuracy was substantiated by an AUC value of 0.8316, confirming an 83.16% accuracy rate.

While effective, the study acknowledges limitations stemming from the scale of the flood inventory utilised and the model’s current inability to simulate real-time flood dynamics. Nevertheless, these findings provide actionable guidance for relevant authorities, aiding in the development of early warning systems, the implementation of protective measures, and the formulation of sustainable zoning policies. This research thus establishes a crucial foundation for effective regional flood risk management.

Data availability

The data and results that support the conclusions of the study are fully available within this article.

Received: 28 November 2025; Accepted: 27 April 2026

Published online: 06 May 2026

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Acknowledgements

This research acknowledges the support provided by the International Partnership Program of Chinese Academy of Sciences (No. 046GJHZ2023071MI) and the Second Tibetan Plateau Scientific Expedition and Research Program (No. 2019QZKK0903).

Author contributions

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Funding

The authors declare that no funds, grants or other support were received during the preparation of this manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

Not applicable. This research did not involve human participants, animal experiments, or any other procedures requiring ethical approval.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-51262-y>.

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